

10-04-2017
BILKENT UNIVERSITY
Department of Electrical and Electronics Engineering
EEE211 Analog Electronics
Midterm Exam Solution

Surname: _____

Name: _____

ID-Number: _____

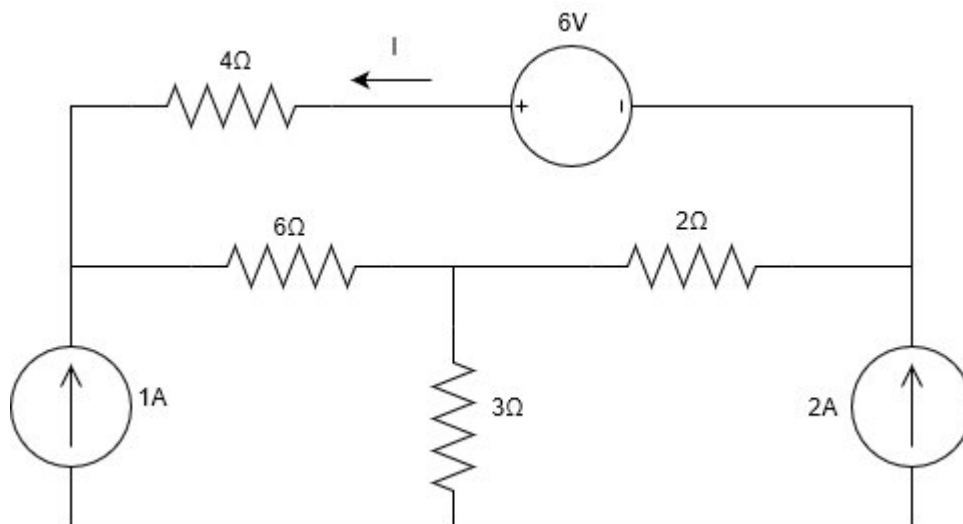
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**Duration is 110 minutes. There are 4 questions of equal weight, solve all.
Show all your work. No books or notes.**

Q1	
Q2	
Q3	
Q4	
Total	

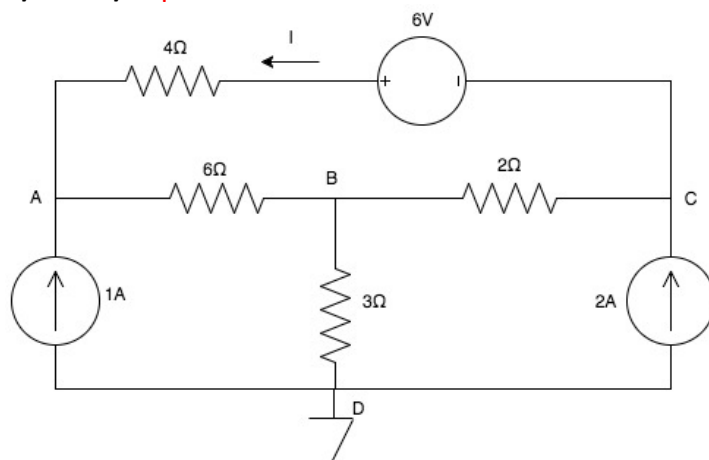
Q1) (25 points) For the circuit below

- Identify and label the nodes as A, B, ...
- Determine a common node
- Write down the node equations in terms of the node voltages
- Find the node voltages and calculate I.



Solution:

a) and b) 2 points



c) (15 points, If A, B are used instead of V_A , V_B in the equations -5)

$$-1 + \frac{V_A - V_B}{6} + \frac{V_A - 6 - V_C}{4} = 0$$

$$\frac{V_B - V_A}{6} + \frac{V_B}{3} + \frac{V_B - V_C}{2} = 0$$

$$\frac{V_C - V_B}{2} - 2 + \frac{V_C + 6 - V_A}{4} = 0$$

$$-12 + 2V_A - 2V_B + 3V_C - 18 - 3V_C = 0 \quad \Rightarrow \quad 5V_A - 2V_B - 3V_C = 30$$

$$V_B - V_A + 2V_B + 3V_B - 3V_C = 0 \quad \Rightarrow \quad -V_A + 6V_B - 3V_C = 0$$

$$2V_C - 2V_B - 8 + V_C + 6 - V_A = 0 \quad \Rightarrow \quad -V_A - 2V_B + 3V_C = 2$$

d)

$$\begin{array}{lll} 5V_A - 2V_B - 3V_C = 30 & 5V_A - 2V_B - 3V_C = 30 & 5V_A - 2V_B - 3V_C = 30 \\ -V_A + 6V_B - 3V_C = 0 & \text{ } \text{ } \text{ } -5V_A + 30V_B - 15V_C = 0 & \text{ } \text{ } \text{ } 28V_B - 18V_C = 30 \\ -V_A - 2V_B + 3V_C = 2 & -5V_A - 10V_B + 15V_C = 10 & -12V_B + 12V_C = 40 \end{array}$$

$$28V_B - 18V_C = 30$$

$$28V_B - 18V_C = 30$$

$$-12V_B + 12V_C = 40 \quad -18V_B + 18V_C = 60 \quad V_C = \frac{60 + 18V_B}{18} = \frac{10}{3} + V_B$$

$$28V_B - 18\left(\frac{10}{3} + V_B\right) = 30 \quad \text{ } \text{ } \text{ } 28V_B - 60 - 18V_B = 30 \quad \text{ } \text{ } \text{ } V_B = 9$$

$$V_C = \frac{10}{3} + 9 = \frac{37}{3} = 12.33$$

$$5V_A - 2V_B - 3V_C = 30 \quad \text{ } \text{ } \text{ } 5V_A - 2 \cdot 9 - 3 \cdot \frac{37}{3} = 30 \quad \text{ } \text{ } \text{ } 5V_A - 18 - 37 = 30$$

$$\text{ } \text{ } \text{ } 5V_A - 18 - 37 = 30 \quad \text{ } \text{ } \text{ } V_A = \frac{85}{5} = 17$$

$$V_A = 17 \quad V_B = 9 \quad V_C = 12.33 \quad \text{(5 points)}$$

$$I = (V_C + 6 - V_A) / 4 = (37/3 + 6 - 17) / 4$$

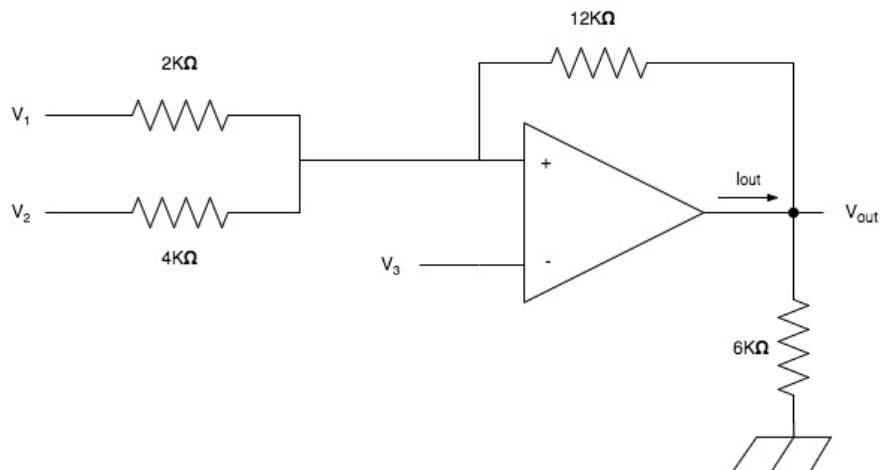
$$= (37/3 - 11) / 4 = (37/3 - 33/3) / 4 = (4/3) / 4 = 0.33 \quad \text{(3 points)}$$

If C is taken as the reference node then $V_A = 14/3 \text{ V}$, $V_B = -10/3 \text{ V}$, $V_D = -37/3 \text{ V}$.

Q2) (25 points) For the circuit below

- Find V_{out} and I_{out} when $V_1 = 12V$, $V_2 = -6V$, and $V_3 = 0$.
- Find V_{out} and I_{out} when $V_1 = 12V$, $V_2 = -6V$, and $V_3 = 5V$.

Note: I_{out} is the current supplied by the opamp.



Solution:

a)

$$V_{out} = -12\left(\frac{V_1}{2} + \frac{V_2}{4}\right) = -12\left(\frac{12}{2} + \frac{-6}{4}\right) = -54V \quad (6 \text{ points})$$

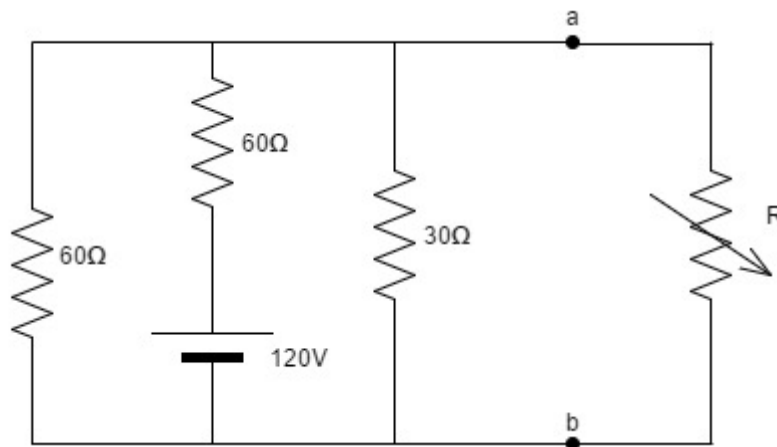
$$I_{out} = \frac{V_{out}}{6} + \frac{V_{out}}{12} = \frac{-54}{6} + \frac{-54}{12} = -13.5mA \quad (4 \text{ points})$$

b)

$$V_{out} = 5 - 12\left(\frac{V_1 - 5}{2} + \frac{V_2 - 5}{4}\right) = 5 - 12\left(\frac{7}{2} + \frac{-11}{4}\right) = -4V \quad (9 \text{ points})$$

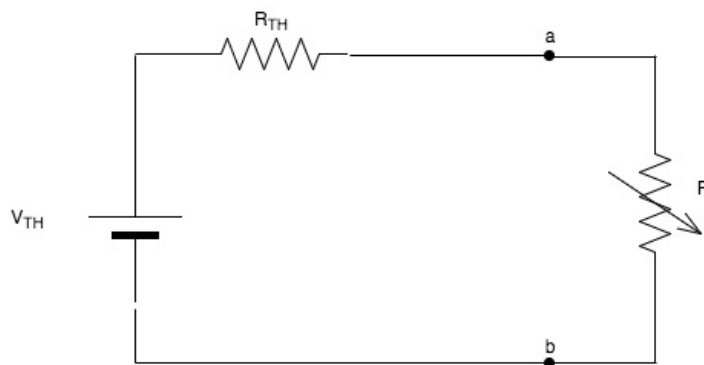
$$I_{out} = \frac{V_{out}}{6} + \frac{V_{out} - 5}{12} = \frac{-4}{6} + \frac{-9}{12} = \frac{-17}{12} = -1.416mA \quad (6 \text{ points})$$

Q3) (25 points) Find the maximum power that the network to the left of terminal a-b can deliver to the adjustable resistor R in the following circuit.
Hint: Finding the equivalent circuit across a-b may be useful.



Solution:

We find the Thevenin equivalent of the circuit to the left of a-b:



$$R_{TH} = 60 // 60 // 30 = 30 // 30 = 15\Omega$$

$$V_{TH} = (60 // 30) / ((60 // 30) + 60) * 120 = (20) / (20 + 60) * 120 = (1/4) * 120 = 30V$$

Maximum power is delivered to R if $R = 15\Omega$.

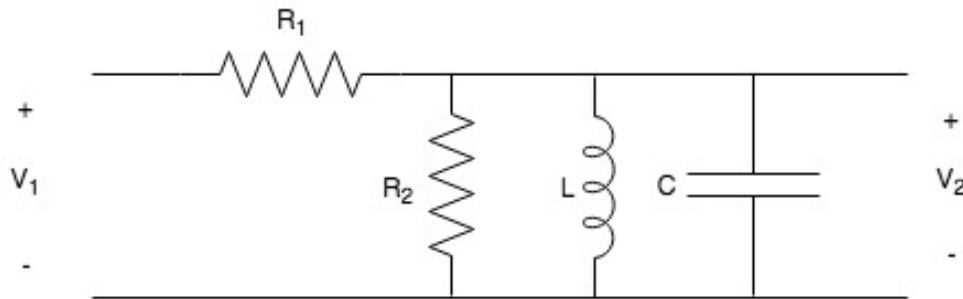
In this case power delivered to the load is $P_{max} = [(30/2)^2] / 15 = 15 * 15 / 15 = 15W$

Q4) (25 points) For the following circuit

a) Obtain the voltage transfer function $H(\omega) = \frac{V_2(\omega)}{V_1(\omega)}$

b) Find the frequency at which $H(\omega)$ is real

c) Find the -3dB bandwidth



Solution:

a)

$$\begin{aligned}
 H(\omega) &= \frac{\frac{1}{\frac{1}{R_2} + j\omega C + \frac{1}{j\omega L}}}{R_1 + \frac{1}{\frac{1}{R_2} + j\omega C + \frac{1}{j\omega L}}} = \frac{1}{1 + R_1 \left[\frac{1}{R_2} + j\omega C + \frac{1}{j\omega L} \right]} = \frac{1}{1 + \frac{R_1}{R_2} + R_1 j \left(\omega C - \frac{1}{\omega L} \right)} \\
 &= \frac{1}{\frac{R_2 + R_1}{R_2} + R_1 j \left(\omega C - \frac{1}{\omega L} \right)} = \frac{1}{\frac{R_2 + R_1}{R_2} \left(1 + \frac{R_1 R_2}{R_2 + R_1} j \left(\omega C - \frac{1}{\omega L} \right) \right)} \\
 &= \left(\frac{R_2}{R_2 + R_1} \right) \frac{1}{\left(1 + (R_1 // R_2) j \left(\omega C - \frac{1}{\omega L} \right) \right)}
 \end{aligned}$$

b) H is real when $\omega C - \frac{1}{\omega L} = 0 \Rightarrow \omega = \frac{1}{\sqrt{LC}}$

c)

$$|H(\omega)|_{\max} = \frac{R_2}{R_2 + R_1}$$

At the -3dB points

$$\left| \left(\frac{R_2}{R_2 + R_1} \right) \frac{1}{\left(1 + (R_1 // R_2) j \left(\omega C - \frac{1}{\omega L} \right) \right)} \right| = \frac{1}{\sqrt{2}} \frac{R_2}{R_2 + R_1}$$

$$\Rightarrow \left| \frac{1}{\left(1 + R_x j \left(\omega C - \frac{1}{\omega L} \right) \right)} \right| = \frac{1}{\sqrt{2}} \quad \text{where } R_x = R_1 // R_2$$

$$\Rightarrow \frac{1}{1 + R_x^2 \left(\omega C - \frac{1}{\omega L} \right)^2} = \frac{1}{2} \Rightarrow R_x^2 \left(\omega C - \frac{1}{\omega L} \right)^2 = 1 \Rightarrow \omega C - \frac{1}{\omega L} = \pm \frac{1}{R_x}$$

$$\omega^2 \mp \frac{w}{CR_x} - \frac{1}{CL} = 0 \Rightarrow \omega^2 \mp \frac{w}{CR_x} - \omega_0^2 = 0 \Rightarrow \omega = \frac{\pm \frac{1}{CR_x} \pm \sqrt{\frac{1}{C^2 R_x^2} + 4\omega_0^2}}{2}$$

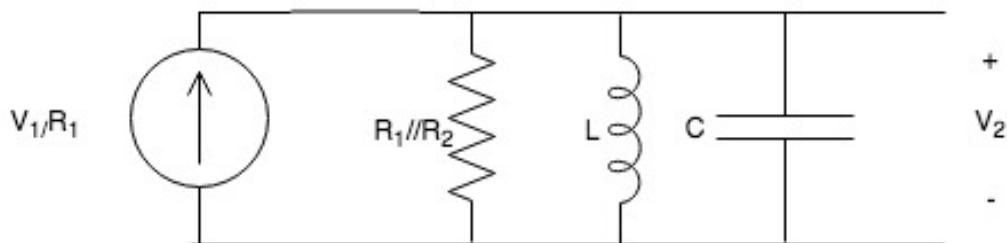
Since frequencies are positive

$$\omega_2 = \frac{+\frac{1}{CR_x} + \sqrt{\frac{1}{C^2 R_x^2} + 4\omega_0^2}}{2} \quad \omega_1 = \frac{-\frac{1}{CR_x} + \sqrt{\frac{1}{C^2 R_x^2} + 4\omega_0^2}}{2}$$

$$BW = \omega_2 - \omega_1 = \frac{1}{2CR_x} - \left(-\frac{1}{2CR_x}\right) = \frac{1}{CR_x} = \frac{R_1 + R_2}{CR_1 R_2}$$

Another solution:

The circuit is equivalent to



This is now a parallel resonant circuit and $Q = \omega_0 R_x C$. Therefore

$$BW = \frac{\omega_0}{Q} = \frac{\omega_0}{\omega_0 CR_x} = \frac{1}{CR_x} = \frac{R_1 + R_2}{CR_1 R_2}$$